

EXPERIMENT 1

Title

Design of ER Diagram for an Indian E-Commerce Platform

Aim

To design an **Entity-Relationship (ER) Diagram** for an Indian e-commerce platform using the entities Customer, Product, Order, OrderItem, Seller, Category, Payment, Delivery, and Address, and to identify primary keys, composite attributes, multi-valued attributes, weak entities, specialization, cardinality, and participation constraints.

Theory

An **ER Diagram** is a graphical representation of a database. It shows the entities, attributes, and relationships between different entities.

In this e-commerce system, the following entities are used:

1. **Customer** – stores customer details.
2. **Address** – stores delivery addresses of customers.
3. **Seller** – stores seller information.
4. **Product** – stores product details.
5. **Category** – stores product categories.
6. **Order** – stores customer order details.
7. **OrderItem** – stores individual products present in an order.
8. **Payment** – stores payment details.
9. **Delivery** – stores delivery and tracking information.

Primary Keys

- Customer → Customer_ID
- Address → Address_ID
- Seller → Seller_ID
- Product → Product_ID
- Category → Category_ID
- Order → Order_ID
- Payment → Payment_ID
- Delivery → Delivery_ID

Composite Key

The **OrderItem** entity has a composite primary key:

(Order_ID, Product_ID)

This is because one order can contain multiple products.

Composite Attribute

The **Address** can be considered a composite attribute consisting of:

House_No + Street + City + State + PIN

Multi-Valued Attribute

A customer can have multiple phone numbers or addresses. Therefore, the customer can have a multi-valued attribute such as:

Phone_Number {multiple values}

In the database design, multiple addresses are represented using the separate **Address** entity.

Weak Entity

OrderItem is treated as a weak entity because it depends on the Order and Product entities for its identification.

Its identification is based on:

Order_ID + Product_ID

Specialization of Product

The **Product** entity is specialized into:

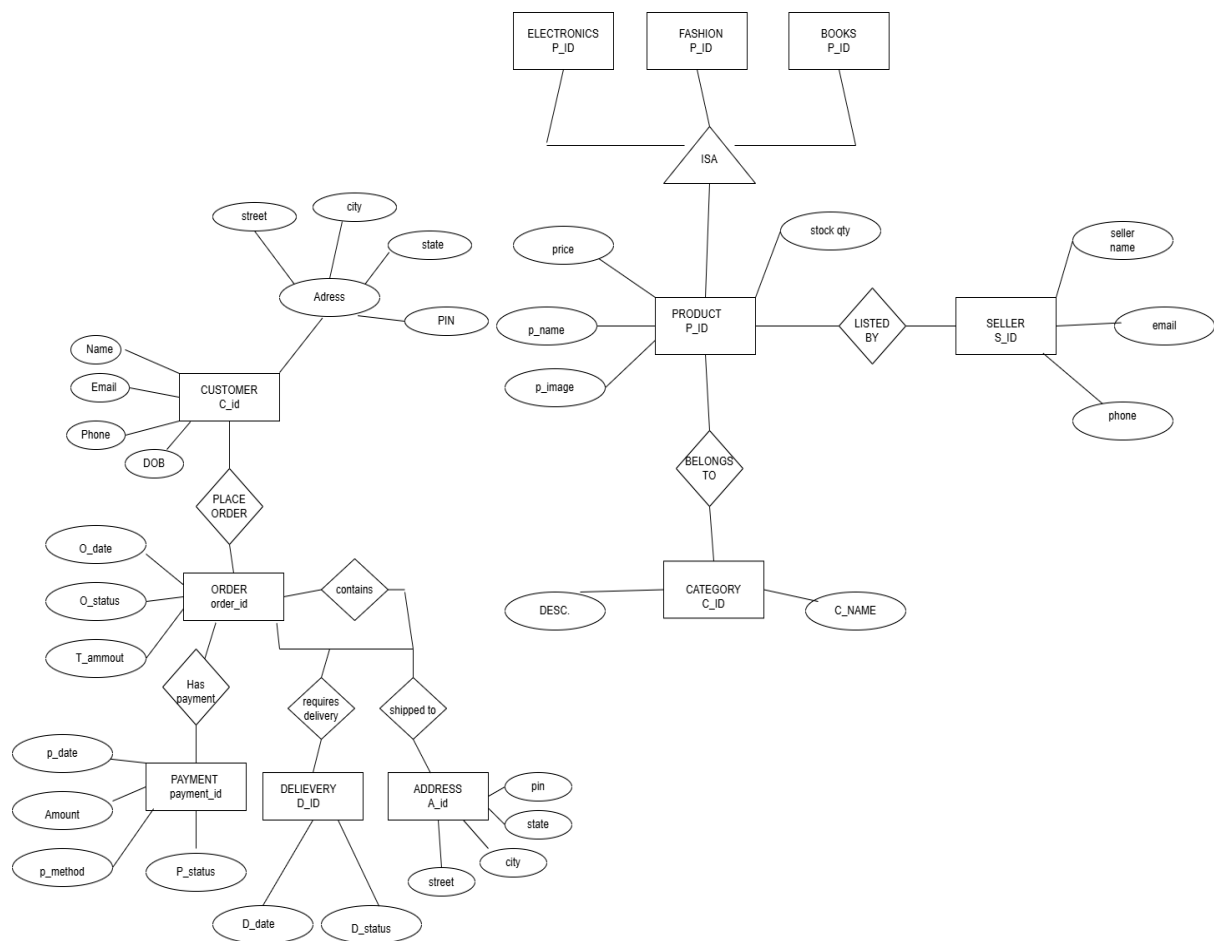
- Electronic
- Clothing
- Grocery

For example:

- Electronic → Brand, Warranty
- Clothing → Size, Material, Gender
- Grocery → Expiry Date, Weight, Food Type

This is called **specialization** or an **ISA relationship**.

Output:-



Result

The ER diagram for the **Indian E-Commerce Platform** was successfully designed. It represents all major entities, their attributes, primary keys, composite key, multi-valued attribute, weak entity, product specialization, relationships, cardinality, and participation constraints.